Mohamed Aamir

PhD Scholar — Ocean Engineering

Indian Institute of Technology Madras

LinkedIn — GitHub — ResearchGate — ORCID

Profile

Sophomore PhD scholar in the Department of Ocean Engineering at IIT Madras, working under the guidance of Dr. K. Narendran. Research interests include hydrodynamics, structural stability, numerical modeling, CFD, and computational analysis of offshore systems.

Education

Indian Institute of Technology Madras — Present
PhD in Ocean Engineering

Mohamed Sathak Engineering College, Kilakarai — 2020 – 2024
B.E. in Civil Engineering
CGPA: 8.56

Syed Ammal Matric Boys Higher Secondary School — 2020
Higher Secondary Certificate (HSC)
Percentage: 83.5%

Syed Ammal Matric Boys Higher Secondary School — 2018
Secondary School Leaving Certificate (SSLC)
Percentage: 97%

Research Focus

Moored Floating Offshore Structures

Research focused on the hydrodynamic response, dynamics, and stability of moored offshore systems under varying sea states.

Teaching Experience

Module Instructor: Introduction to OpenFOAM — May 2025

Conducted the OpenFOAM module for *OE5555: Summer Training in Computer Modeling and Simulation*. Instructed Master's students on CFD workflows, mesh generation, solver setup, and hydrodynamic simulations.

Internship Experience

Structural Engineering Intern — Jan – May 2024
Almino Structural Consultancy, Ramanathapuram

- Assisted in structural drafting and detailing using AutoCAD
- Performed rebar checking and site verification activities

Selected Coursework

- Numerical Techniques in Ocean Hydrodynamics
- Experimental Methods and Measurements
- Installation of Offshore Structures
- Wave Hydrodynamics
- Machine Learning for Ocean Engineers
- Pipeline & Riser Engineering
- Coastal Engineering
- Advanced Dynamics of Floating Bodies

Technical Skills

- OpenFOAM
- CFD and Numerical Modeling
- Hydrodynamic Analysis
- MATLAB
- Python
- AutoCAD

Research Outputs

Manuscripts and research outputs currently under preparation.